FIITJEE

ALL INDIA TEST SERIES

FULL TEST – III

JEE (Advanced)-2021

PAPER –2

TEST DATE: 18-01-2021

Time Allotted: 3 Hours **Maximum Marks: 198**

General Instructions:

- The test consists of total **54** questions.
- Each subject (PCM) has **18** questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections**: **Section-A, Section – B & Section-C.**

Section-A (01 – 06, 19 – 24, 37– 42) *this section contains* ***18 multiple*** *choice questions.*
Each question has FOUR options. ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct answer(s).*
For each question, choose the option(s) corresponding to (all) the correct answer(s)
Answer to each question will be evaluated according to the following marking scheme:

Full Marks **: +4** If only (all) the correct option(s) is (are) chosen:
Partial Marks **: +3** If all the four options are correct but ONLY three options are chosen;
Partial Marks **: +2** If three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks **: +1** If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks **: 0** If none of the options is chosen (i. e. the question is unanswered);
Negative Marks **: –2** In all other cases

Section-B (07 – 12, 25 – 30, 43– 48) *contains* ***18 Numerical*** *based questions with* ***Single digit integer*** *as answer, ranging from* ***0 to 9*** *and each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Section-C (13 – 18, 31 – 36, 49– 54) *contains* ***18 Numerical*** *answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer and* ***0 marks*** *for wrong answer.*

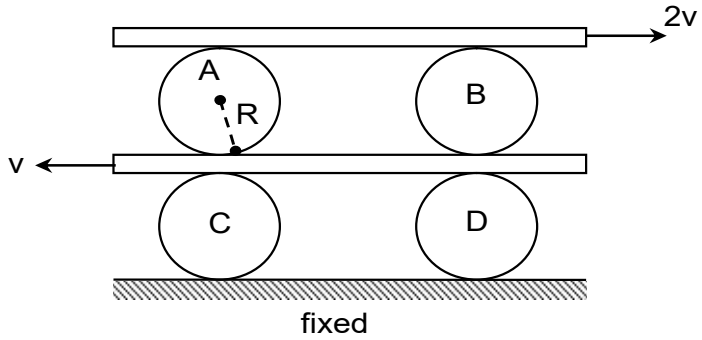
Physics

PART – I

SECTION – A
(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. Two thin planks are moving on four identical solid cylinders of mass m and radius 'R' as shown in figure. There is no slipping at 2v any contact choose the correct option.



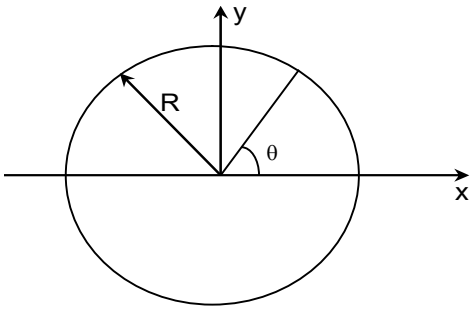
(A) kinetic energy of cylinder 'A' is $\frac{11}{16}mv^2$

(B) kinetic energy of cylinder C is $\frac{3}{8}mv^2$

(C) kinetic energy of cylinder C is $\frac{3}{5}mv^2$

(D) kinetic energy of cylinder D is $\frac{3}{16}mv^2$

2. On a non conducting ring of radius R linear charge density 'λ' given by $\lambda = \lambda_0 \cos\theta$ is distributed. Choose correct option(s).



(A) Electric potential at centre is zero

(B) Electric field at centre is zero

(C) Electric field at centre is $\frac{\lambda_0}{4\varepsilon_0}$

(D) Dipole moment of ring is $\lambda_0 \pi R^2$

3. One mole of ideal monoatomic gas is expanded to double volume, through a process $\frac{P^2}{\rho} =$ constant. Initial temperature is T, pressure is P and density is ρ. Choose the correct option.

(A) Temperature of gas changes to $\sqrt{2}$ T.

(B) P – T diagram of the process is hyperbola

(C) Work done by the gas is $2\left(\sqrt{2}-1\right)R$

(D) Internal energy of gas changes by $\frac{3}{2}RT$

4. An elastic spherical ball of mass m and radius 'a', moving with speed v, strikes a fixed rough horizontal surface at an angle θ with normal. After striking it rebounds at an angle ϕ with normal. If coefficient of friction is small enough to prevent skidding. Then which of the following may be correct.

(A) $\tan\phi = \tan\theta + 2\mu$

(B) change in angular speed is $\frac{5\mu v\cos\theta}{a}$

(C) $\tan\theta = \tan\theta - 2\mu$

(D) change in angular speed is $\frac{5\mu v\cos\theta}{7a}$

5. A uniformly charged small sphere is placed at point A (h, k) so that at point B (8, 7) electric field strength is $\vec{E} = \left(54\ \hat{i} + 72\ \hat{j}\right) N/C$ and potential is + 900 volt. Then choose the correct option(s).

(A) charge on the sphere is $1\ \mu C$

(B) charge on the sphere is $4\ \mu C$

(C) co-ordinates of point A is $(2,-1)$

(D) co-ordinates of point A is $(4,-2)$

6. A thin ring of linear mass density 'λ' and radius 'R' having cross sectional area 'a' is rotating with angular speed 'ω'. If Y is young modulus of ring, then choose the correct option(s).

(A) Kinetic energy of ring is $\pi R^3\lambda\omega^2$

(B) Tension in the ring is $\frac{\lambda\omega^2R^2}{2}$

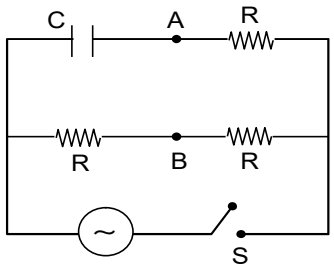
(C) Elastic potential energy stored in ring is $\frac{2\pi\lambda^2\omega^4R^5}{Y.a}$

(D) All of these

SECTION – B
(Single Digit Integer Type)

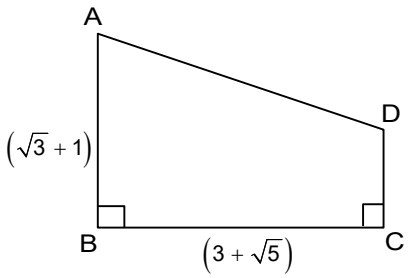
This section contains **06** questions. The answer to each question is a **Single Digit integer** ranging from **0** to **9**, both inclusive.

7. Consider a circuit shown in figure switch was open for a long time and closed at t = 0. All resistors are of resistance $R = 50\ \Omega$. What is maximum value of potential difference between points A & B? (Given C = 100 μF)



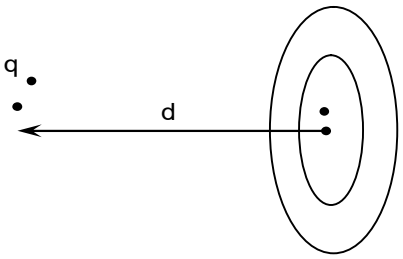
V = 1/8 sin(100t)

8. Consider a uniform plate in the form of a trapezium. When this plate is suspended vertically from corner 'D'. It is found that side BC remains horizontal in equilibrium. Find side length 'CD'



9. A uniform disc of mass m and radius R having charge Q distributed uniformly all over its surface is placed an a smooth horizontal surface. A magnetic field $B = B_0 rt^3$, where B_o is constant, r is radial distance from the centre of disc and t is time is switched on perpendicular to plane of disc. Angular acceleration of disc at time t is given by $p\dfrac{B_0 QRt}{15m}$. Find p?

10. A charge q_o is placed at a distance of d = 12 cm from the centre of hollow disc of inner radius 9 cm and outer radius 16 cm. If flux of electric field through the disc area is $\dfrac{n.q_0}{5\in_0}$. Find n?



11. A particle of mass m is projected from surface of planet of mass $M(>> m)$ with speed $\sqrt{\dfrac{7GM}{4R}}$. tangential to the surface of planet and it goes maximum height h' from the surface. Find h/R (R is radius of planet)

12. When a soap bubble of radius R = 0.25 cm is charged with surface charge density $\sigma = 20\ \mu C/m^2$. If in this situation pressure inside and outside the bubble is equal surface tension of liquid is given by $N\times 10^{-12}/8\in_0$ N/M. Find N.

SECTION – C
(Numerical Answer Type)

This section contains **06 questions**. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **SECOND DECIMAL PLACE**; e.g. XXXXX. XX).

13. A wheel of radius 20 cm is in pure rolling on a 'rough horizontal surface'. Centre of wheel is moving with v = 4 m/s consider two end points on vertical diameter. What is relative acceleration of one with respect to second point. (in m/s^2)

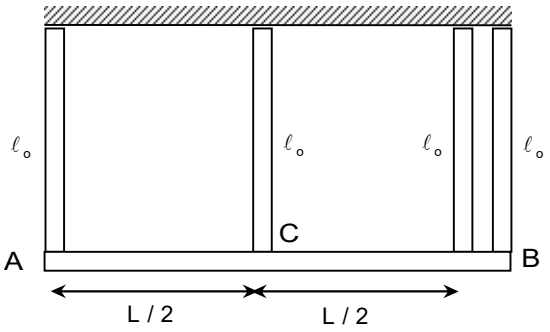
14. A cylindrical conductor of length L and area A has variable electrical resistivity $\sigma = \sigma_0(1+x/L)$. where x is distance from one end. The end at x = 0 is given a potential V_o where and at x = L is grounded. If electric field at x = L/2 is given by $E = \dfrac{V_0}{\alpha L.\ell n\beta}$. Find $(\alpha+\beta)$

15. A disc of moment of Inertia 24 kg m^2 is rotating about its axis with angular speed 15 rad/sec. If axis is also rotating with angular speed of 6 rad/sec. Find the torque exerted by axis on the disc (in Nm)

16. A nucleus X initially at rest, undergoes $\alpha-$decay according to the equation ${}^{225}_{92}X \rightarrow Y + \alpha$
The emitted $\alpha-$particle is found to move along a helical path in a uniform magnetic field of induction B = 5 T, Radius and pitch of the helix traced by the $\alpha-$ particle are R = 5 cm and $p = 7.5\pi$ cm calculate the binding energy per nucleon of nucleus X (in MeV) (Given m(v) = 221.03 u, $M(\alpha) = 4.003$ u, $M(n) = 1.009$ u, $M(p) = 1.008$ u)

17. A small metallic sphere of density three times that of ice radius 4.2 cm and specific heat capacity 500 j/kg^{o}C is heated to temperature 800° C and then placed over large block of ice. Maximum depth to which sphere can penetrate ice (in cm) is

18. A uniform rod AB of mass M and length L is suspended from four identical wires with two wires at end B, one at end A and one at centre. Natural length of wires is ℓ_0. Tensions in wires connected at A is given by $\frac{nT}{3n-1}$. Find n.



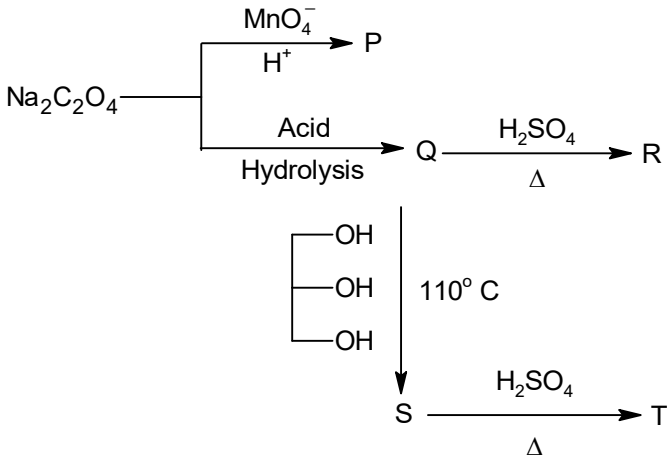
Chemistry

PART – II

SECTION – A
(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

19. Consider the following reaction,



(A) P can be carbon dioxide
(B) R can be carbon dioxide
(C) R can be carbon monoxide
(D) T can be carbon monoxide

20. Which of the statement(s) are true?
(A) The equivalent weight of Na_3PO_4 is Mw/3
(B) The highest percentage labelling on oleum bottle is 122.5
(C) The oxidation state of iron in $\left[Fe(NO)(H_2O)_5\right]SO_4$ is + 1
(D) The average oxidation number of iron in Prussian blue is 18/7

21. Which of the following statement/s are incorrect?
(A) Cisplatin is carcinogenic agent
(B) Dinitrogen trioxide, a black solid, is anhydride of nitrous acid
(C) Liquid helium is used as cryogenic agent
(D) Neon is used in fluorescent bulbs for advertisement display.

22. Select the incorrect statement(s)?
(A) The kinetic energy of 7.5 g molecules of ethane is $\frac{3RT}{8}$
(B) The volume of acetylene required for producing 7.8 g benzene is $\frac{3RT}{10\ P}$, at given temperature T
(C) Pressure correction term in van der Waal's equation for n moles of gas $= \frac{an^2}{V^2}$
(D) Hydrogen will diffuse 1.46 times faster than ammonia gas.

23. Select the correct statement(s):
(A) Sucrose contains $\beta-2,1$ glycosidic linkage
(B) Sucrose contains $\alpha-1,2$ glycosidic linkage
(C) Maltose contains $\alpha-1,4$ glycosidic linkage
(D) Lactose contains $\beta-1,\ 4$ glycosidic linkage

24. In which of the following reaction, given change does not takes place?
(A) Formic acid $\xrightarrow[H_2SO_4]{conc}$ Decarboxylation
(B) Oxalic acid $\xrightarrow[H_2SO_4]{conc}$ Decarboxylation
(C) $\beta-$keto acid $\xrightarrow{\Delta}$ Decarboxylation
(D) Succinic acid $\xrightarrow{\Delta}$ Decarboxylation

SECTION – B
(Single Digit Integer Type)

This section contains **06** questions. The answer to each question is a **Single Digit integer** ranging from **0** to **9**, both inclusive.

25. 250 mL of a solution containing 0.6625 g of sodium carbonate and 2.8000 g of potassium hydroxide is titrated against N/10 HCl. Find out the value of x – y, where x and y are titre value (in mL) with methyl orange and phenolphthalein, respectively, taking 20 mL of above solution each time.

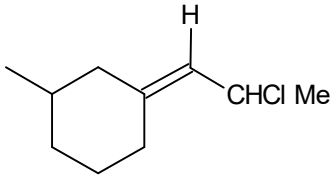
26. The phenomenon of emission of electrons from surface of a metal on irradiation with light of suitable frequency is known as photo electric effect. Now, calculate x with the help of following data for a certain metal.

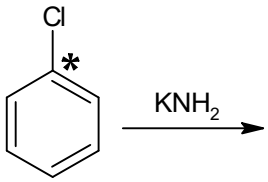
Incident light	K.E. of photoelectrons	Threshold frequency
(i) 2×10^{15} Hz	KE	$x\times10^{-15}$
(ii) 5×10^{15} Hz	4KE	$x\times10^{-15}$

27. Among the molecules/ions, $P(CH_3)_3(CF_3)_2$, SF_4, $[PtCl_4]^{2-}$, XeF_4, I_2Cl_6, SF_6, IF_7, $[Pt(NH_3)_6]^{+4}$, $[XeOF_5]^-$, the total number of molecules/ions containing ≥ 5 atoms in one plane, is

28. $$H-(CH_2)_n-\overset{\overset{O}{\|}}{C}-OH \xrightarrow[HVZ]{Cl_2/P} \text{Product/s}$$
Total number of products (including stereoisomer, if any) formed in above reactions for n = 0, 1, 2 is

29. Calculate the total number of optical isomer possible for



30. 

Total number of products formed in above reaction is

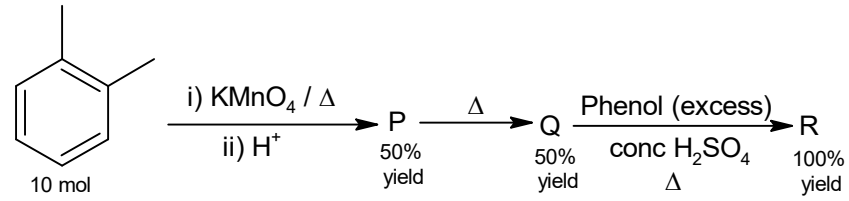
SECTION – C
(Numerical Answer Type)

This section contains **06 questions**. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **SECOND DECIMAL PLACE**; e.g. XXXXX. XX).

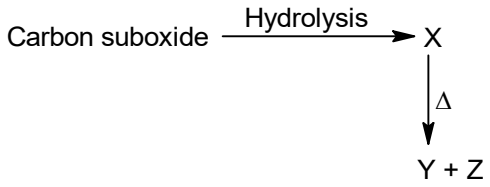
31. The pH of 0.01 M aqueous solution of aliphatic amine $(R-NH_2)$ is 11. If value of K_b is $x\times10^{-4}$, then find the value of 5x.

32. Consider wurtzite form of zinc sulphide where sulphide ion forms the one hexagonal unit cell. If the number of zinc ion and sulphide ion, completely inside the single unit cell, are x and y respectively, then calculate the value of x/y.

33. 0.1 molal aqueous solution of a weak acid H – A freezes at – 0.0223°C. Calculate the pH of above solution, assuming molarity = molality. (Given $K_f = 0.1860$, $\log 2 = 0.3$, At. mass of H = 1, C = 12, O = 16)

34. 

The amount of R produced in gram is

35. 

The number of gram equivalents of $\overline{O}H$ required for complete neutralization of mixture containing 1.1×10^3, 2.2×10^3 and 2.2×10^3 moles of X, Y and Z respectively, is

36. Calculate the maximum number of moles of gas produced in below three reactions.

(i) $N_{2(g)} + H_{2(g)} \longrightarrow NH_{3(g)}$

310.8 Kg 22.2 Kg

(ii) $CaCO_{3(s)} \xrightarrow{\Delta} CO_{2(g)} + CaO_{(s)}$

100 Kg

(iii) $NH_4NO_{3(s)} \xrightarrow{\Delta} H_2O_{(\ell)} + N_2O_{(g)}$

80 Kg

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. If $f^3(x)-3x^2f(x)+x^3=0$, then $\int\frac{f(x)}{x}dx$ is equal to

(A) $f(x)+c$

(B) $-f(x)+c$

(C) $2f(x)+c$

(D) $xf'(x)+c$

38. If *f* is continuous and one-one function for $x\geq 0$ and $\int_0^x f(t)dt<\frac{x}{2}\left(f(0)+f(x)\right)$, then correct option may be

(A) $f'(x)<0;\ f''(x)<0$

(B) $f'(x)<0;\ f''(x)>0$

(C) $f'(x)>0;\ f''(x)<0$

(D) $f'(x)>0;\ f''(x)>0$

39. If $A=\begin{pmatrix}\alpha & 0\\ 2 & \alpha\end{pmatrix}$ and $\det\left(2A^2-2A\right)=144$, then possible correct statement is (for real part)

(A) $\text{tr}(A)=6$

(B) $\det(A)=9$

(C) $\det(A)=4$

(D) $\text{tr}(A)=1$

40. If $f(x)=\begin{cases}-x^2, & x\geq 0\\ (a^2-2a)+(b^2-2)x, & x<0\end{cases}$ and $x=0$ is point of maxima, then

(A) $a\in(0,\ 2)$ for $b\in\left(-\sqrt{2},\ \sqrt{2}\right)$

(B) $a\in[0,\ 2]$ for $b\in R-\left(-\sqrt{2},\ \sqrt{2}\right)$

(C) $a\in\{0,\ 2\}$ for $b\in R-\left(-\sqrt{2},\ \sqrt{2}\right)$

(D) $a\in R-(0,\ 2)$ for $b\in R-\left[-\sqrt{2},\ \sqrt{2}\right]$

41. If f(x) is a polynomial of degree three with leading coefficient as unity s.t f(1) = 1; f(3) = 9 & f'(1) = 2, then
(A) f'(x) = 2x has at least one between (1,3)
(B) f''(x) = 2 has at least one root between (1, 7/3)
(C) $\frac{d}{dx}\left(\frac{f(x)-x^3}{(x-1)^2}\right)$ at x = 1 is equal to 1
(D) sum of root of f(x) = 0 is 3

42. If f(x) = $\frac{x+\left[1-f^2(x)\right]}{1+x^2}$ is an odd function where [.] denotes greatest integer part, then:
(A) f(x) is invertible for [-1,1]
(B) $f(x) = f^{-1}(x)$ has no solution $-1 \le x \le 1$.
(C) $|f(x)| = K$ has four solution for $0 < K < 1$.
(D) f(x) is differentiable for all R

SECTION – B
(Single Digit Integer Type)

This section contains **06** questions. The answer to each question is a **Single Digit integer** ranging from **0** to **9**, both inclusive.

43. If $f(x) = \left\{b^2 + (a-1)b + 2\right\}x - \int\left(\sin^2 x + \cos^4 x\right)dx$ be an increasing function of $x \in R$ and $b \in R$, then sum of all possible integral values of a is

44. If $I = \int_1^2 \frac{x^3 - 3x^2}{\ln(x-1)}dx$ and $J = \int_2^3 \frac{(3x-5)}{\ln(x-2)}dx$, then e^{I+J} equals …….

45. Position vector of a point A is $\left(a\hat{i} + b\hat{j} + c\hat{k}\right)$ and $a^2 + b^2 + c^2 = 4$; and position vector of another point B is $\left(p\hat{i} + q\hat{j} + r\hat{k}\right)$ and $p^2 + q^2 + r^2 = 9$. Given $\angle AOB = 2\cos^{-1}\frac{1}{5}$, where O is origin. If AQ is perpendicular to angle bisector of $\angle AOB$ i.e. line OQD where D lies on AB, then value of $25|QD|$ = ?

46. If $\frac{3x^2+1}{\sqrt{x^4+x^2}} \ge f(x)$, where $f(x)$ is a constant function in $0 < x \le \sqrt{2}$, then minimum value of $\int_0^{\sqrt{2}} \frac{(3x^2+1)dx}{\sqrt{x^4+x^2}}$ is ……

47. If $\left|\text{adj}\left(\text{adj}\left(\text{adj } A\right)\right)\right| = \left|\text{adj } (A)\right|^4$, then number of entries in diagonal of *A* is ……..

48. If $AB = BA = I$ such that $A = \left[a_{ij}\right]_{2\times 2}$, $B = \left[b_{ij}\right]_{2\times 2}$, $a_{ii} = b_{ii}$, $a_{12} = -b_{12}$, $a_{21} = -b_{21}$ and a_{ij} are co-prime natural numbers, then $|a_{21} - a_{12}| =$ …….

SECTION – C
(Numerical Answer Type)

This section contains **06 questions**. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **SECOND DECIMAL PLACE**; e.g. XXXXX. XX).

49. If $P_n = 1+\cos 2^{n-1}\theta - \sin 2^{n-1}\theta$ and $Q_n = 1+\tan 2^{n-1}\theta$ such that $(P_1Q_0)\times(P_2Q_1)\times.....(P_nQ_{n-1}) = 1$ for $n \to \infty$, then value of $\frac{\pi}{\theta.2^{n-1}}$ is

50. If $\vec{u}(x) = x\hat{i} + (\sin x)\hat{j} + (1+\sin x)\hat{k}$, $\vec{v}(x) = \hat{i} + \cos x\left(\hat{j}+\hat{k}\right)$ and $\vec{w}(x) = \sin x\left(\hat{j}+\hat{k}\right)$, all are linearly dependent vectors for $x = \alpha$, then $\left(\frac{2\alpha}{\pi}\right)^2$ is divided by 8 leaves remainder …….

51. If $f(x) = (\cos x)\hat{i} + (\sin x)\hat{j} + (\cos 2x)\hat{k}$, $g(x) = (\tan x)\hat{i} + (\sin 3x)\hat{j} + (\cos 4x)\hat{k}$ and $h(x) = (\cos 3x)\hat{i} + (\sin 5x)\hat{j} + (\cos 6x)\hat{k}$ are three variable vectors given, then $\int_0^{\pi}\left[f(x),\ g(x),\ h(x)\right]dx$, where [.] denotes scalar triple product, is equal to

52. If $\vec{a}_1 = a\hat{i} + b\hat{j}$, $\vec{a}_2 = \vec{a}_1 - \left(2\hat{i} + \sqrt{5}\hat{j}\right)$ and $\vec{a}_3 = \vec{a}_1 + \left(2\hat{i} - \sqrt{5}\hat{j}\right)$, then $\begin{vmatrix} 1+\vec{a}_1.\vec{a}_1 & 1+\vec{a}_1.\vec{a}_2 & 1+\vec{a}_1.\vec{a}_3 \\ 1+\vec{a}_2.\vec{a}_2 & 1+\vec{a}_2.\vec{a}_2 & 1+\vec{a}_2.\vec{a}_3 \\ 1+\vec{a}_3.\vec{a}_1 & 1+\vec{a}_3.\vec{a}_2 & 1+\vec{a}_3.\vec{a}_3 \end{vmatrix}$ is equal to

53. If $|f(x)| \le |\sin \pi x|$ and $g(x) = f'(x)$, then maximum value of $|g(n)|$ for $n \in I$ is

54. If $f(x) = \frac{\alpha x+1}{x-\beta}\ \forall x \in R - \{\beta\}$ & $\alpha\beta \ne 1$ & $\alpha \ne 1$ is a self – inverse function such that $\frac{f(4)}{4} = \frac{f(12)}{12} = f\left(\frac{1+\beta}{1-\alpha}\right)$, then "β" is equal to